# ColRadPy

## Theory Manual

Collisional radiative kinetics and spectral emission

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This manual develops the rate equations, the elimination of excited states, generalized collisional radiative coefficients, time-dependent and charge-state solutions, line emission, and the energy-balance extension. Equations are mapped to the routines that implement them. Blue notes explain implementation choices; orange notes identify current limitations or qualifications. The worked checks execute the development working tree; file hashes and the recorded HEAD are given in `docs/source/theory/figures/provenance.json`.

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# 1 Purpose and model scope

ColRadPy converts level-resolved atomic data into populations, photon emission coefficients, and effective rates for charge-state evolution. The basic input is an ADAS ADF04 description of one ion: level energies, radiative probabilities, electron-impact collision strengths, and any supplied parent-resolved ionization, recombination, or charge-exchange data. The project paper introduces this generalized collisional radiative (GCR) approach and compares selected results with ADAS calculations [1]. Implementation statements throughout this manual refer to the development working tree used to generate its figures and checks [5].

The plasma parameters are prescribed. Electrons are Maxwellian at a specified temperature, atomic processes are represented by rate coefficients, and emitted photons escape. Ground and metastable populations can be retained as independent drivers. The principal calculation does not solve radiative transfer, a spatial transport equation, or the electron distribution function. Its rate coefficients can instead be passed to a transport or plasma model.

The collisional radiative regime lies between two useful limits. At sufficiently low density, excitation is balanced mainly by radiative decay, giving the coronal limit. At high density, reversible collisions tend toward a Boltzmann distribution within a charge stage; ionization and three-body recombination also tend toward Saha balance when their inverse processes and statistical weights are consistent. These are limits to test, not assumptions that every fusion or astrophysical plasma automatically satisfies.

## 1.1 Two calculations with different meanings

The quasistatic calculation sets derivatives of selected excited populations to zero while retaining the ground and metastables as drivers. It produces linear response coefficients. The fully time-dependent calculation propagates the entire supplied rate matrix from an initial vector. A separate ionization-balance calculation joins effective coefficients from adjacent charge stages. Keeping these distinctions prevents an excited-state response per parent from being mistaken for a normalized ion fraction.

**Operating status.** The development implementation uses current NumPy membership and integration APIs, guarded matrix-exponential propagation, and density-scaled XCD. The worked checks use the runtime recorded in the provenance file. Passing these checks does not validate every atomic dataset or extrapolation regime.

## 1.2 Calculation flow

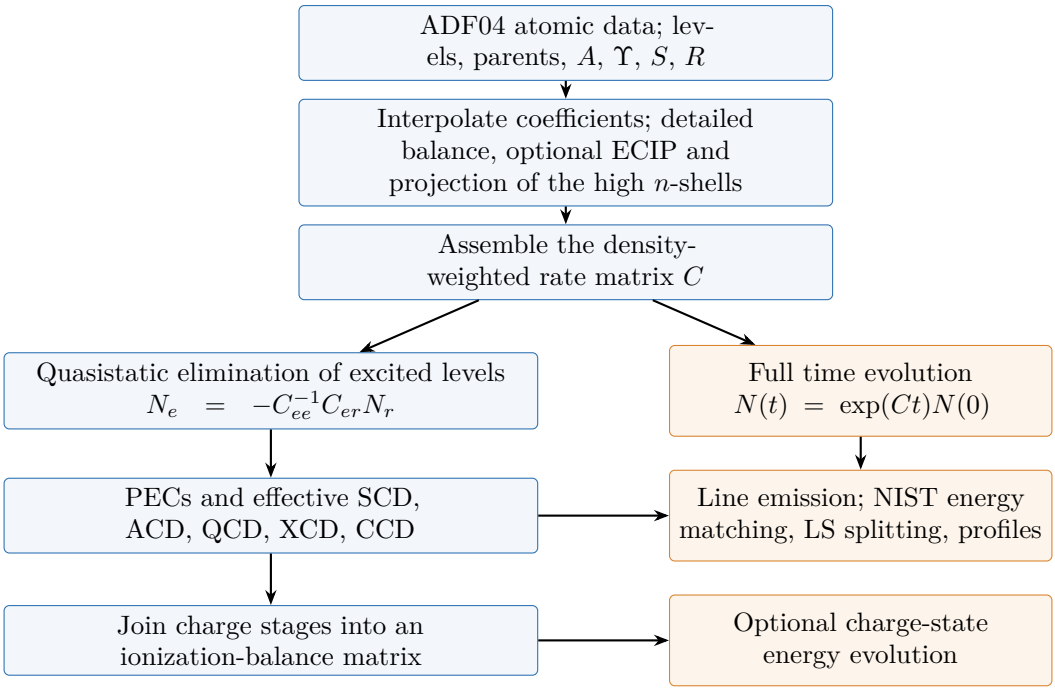



**Figure 1:** The calculation paths are related but do not share the same population normalization. The full matrix can also include parent or reservoir states; their interpretation must be specified.

## 2 Notation, indexing and units

Let $N_i^{(z)}$ denote the number density of level $i$ in charge state $z$. Indices $i, j$ generally label bound levels; $\rho, \sigma$ label retained ground or metastable levels; $p, q$ label parents in charge state $z+1$. The subscripts $e$ and $r$ on matrix blocks denote eliminated excited and retained states, respectively. They do not denote electron density.

The defining matrix convention is

$$\frac{\mathrm{d}N_i}{\mathrm{d}t} = \sum_j C_{ij} N_j, \qquad C_{ij} : \; j \longrightarrow i. \tag{2.1}$$

Thus a column is a source state and a row is a destination. Off-diagonal rates are nonnegative for a physical model. This convention differs from some stored coefficient arrays, as listed in Section 9.

The conversions needed repeatedly are

$$\Delta E_{\mathrm{eV}} = \frac{\Delta E_{\mathrm{cm}^{-1}}}{8065.54}, \qquad T_{\mathrm{K}} \simeq 11604.5\, T_{\mathrm{eV}}, \tag{2.2}$$

$$\lambda_{\mathrm{vac,nm}} = \frac{10^7}{|E_u - E_\ell|_{\mathrm{cm}^{-1}}}, \qquad n_{\mathrm{m}^{-3}} = 10^6 n_{\mathrm{cm}^{-3}}. \tag{2.3}$$

The code uses several slightly different rounded energy-conversion constants. For reproducibility, the exact expressions in a routine take precedence over a claim that all those constants are identical.

ADF04 level and parent identifiers are one-based. Python level indices are zero-based; `metas=[0,2]` retains the first and third bound levels. The complement is then $[1, 3, 4, \ldots]$, not the contiguous interval beginning at the number of metastables. A parent identifier is also not interchangeable with the position at which that identifier was discovered in a file.

| Quantity | Meaning | Units |
|---|---|---|
| $T_e, T_i$ | Temperature represented as an energy | eV |
| $E_i,\ I_p$ | Tabulated level energies and parent thresholds | $cm^{-1}$ |
| $n_e, n_h, N_i$ | Electron, neutral-donor and level densities | $cm^{-3}$ |
| $A_{u\ell}$ | Spontaneous emission probability | $s^{-1}$ |
| $q, S, \alpha^{RR+DR}, K$ | Two-body coefficients | $cm^3\,s^{-1}$ |
| $\alpha^{(3)}$ | Three-body recombination coefficient | $cm^6\,s^{-1}$ |
| $C$ | Density-weighted rate matrix | $s^{-1}$ |
| $\Upsilon$ | Effective collision strength | dimensionless |
| $\lambda$ | Internal wavelength | nm |
| PEC | Photon emission coefficient | $cm^3\,s^{-1}$ |
| SCD, ACD, QCD, XCD, CCD | Effective two-body-form coefficients | $cm^3\,s^{-1}$ |

**Table 1:** Core conventions. Effective coefficients may themselves depend on density. The energy-transfer helper functions use SI densities and masses.

# 3 Atomic data and rate coefficients

## 3.1 Level structure and statistical weights

The ADF04 reader stores configurations, multiplicities, orbital labels, energies, and a weight-related field `atomic['w']`. The matrix builder uses $g_i = 2w_i + 1$. In an LS-resolved file the level weight is $(2S+1)(2L+1)$; in a resolved-$J$ description it is $2J+1$. The parent term must be interpreted consistently with that resolution. The header provides parent ionization thresholds, while level annotations describe parentage channels. They need not enumerate identical sets of observed pathways [3].

**Parent weights.** parent_statistical_weight parses LS and resolved-$J$ parent terms, including fractional $J$, and uses $g_p = (2S+1)(2L+1)$ or $g_p = 2J+1$ as appropriate. Header parent counts and discovered pathway IDs still require care: an entry can have explicit ionization data even when its parentage annotations are incomplete.

## 3.2 Maxwellian electron-impact excitation

For an electron-energy distribution $f_E(E)$ normalized to unity,

$$q_{\ell u} = \int_{\Delta E}^{\infty} \sigma_{\ell u}(E) \sqrt{\frac{2E}{m_e}}\, f_E(E)\, \mathrm{d}E. \tag{3.1}$$

The tabulated effective collision strength $\Upsilon_{\ell u}(T)$ packages the Maxwellian average in a convenient dimensionless form. The implemented rate conversion for $u > \ell$ is

$$q_{u\ell}(T_e) = \frac{2.1716 \times 10^{-8}}{g_u} \sqrt{\frac{13.6058}{T_e}}\, \Upsilon_{\ell u}(T_e), \tag{3.2}$$

$$q_{\ell u}(T_e) = \frac{g_u}{g_\ell} \exp\left(-\frac{\Delta E}{T_e}\right) q_{u\ell}(T_e). \tag{3.3}$$

Here $\Delta E$ and $T_e$ have the same energy units. Both coefficients are in $cm^3\,s^{-1}$. Their ratio enforces collisional detailed balance: $N_u/N_\ell = (g_u/g_\ell)\exp(-\Delta E/T_e)$ when collisions dominate and no other mechanism fixes the ratio.

**Implementation.** make_electron_excitation_rates interpolates $\Upsilon$ onto the user grid. populate_cr_matrix constructs the upward and downward rates and multiplies them by electron density. The excitation, ionization and three-body Boltzmann factors convert $\mathrm{cm}^{-1}$ to eV with $hc/e$ (1/8065.544), so excitation alone drives the populations to the Boltzmann distribution at high density; the ECIP exponent keeps the constants of r8ecip.

## 3.3 Spontaneous decay, ionization and recombination

For a radiative transition $u \to \ell$, the population-transfer rate is $A_{u\ell}N_u$, with no electron-density multiplier. The total radiative loss is $\Gamma_u = \sum_{\ell<u} A_{u\ell}$.

Let $S_{ip}$ denote ionization from bound level $i$ to parent $p$. The reduced ionization quantity in the supported ADF04 convention is converted as

$$S_{ip}(T_\mathrm{e}) = \widetilde{S}_{ip}(T_\mathrm{e}) \exp[-\chi_{ip}/T_\mathrm{e}], \qquad \chi_{ip} = I_p - E_i. \tag{3.4}$$

An interpolated reduced coefficient is not itself the final ionization rate. The input parentage factor can be applied through `scale_file_ioniz`, which multiplies the interpolated reduced coefficients whatever the interpolation; the ionization balance passes the option to each stage. Some files hold placeholders instead of rates at their lowest temperatures, where the rate underflowed when the file was made (reduced ionization rates of $10^{-2}$ to $10^{40}\,\mathrm{cm}^3\,\mathrm{s}^{-1}$ in OpenADAS neon files). Reduced ionization rates above $10^{-6}\,\mathrm{cm}^3\,\mathrm{s}^{-1}$ and recombination rates above $10^{-7}\,\mathrm{cm}^3\,\mathrm{s}^{-1}$, far above physical values, are left out with a warning, and those temperatures take the rates from the other points of the transition. Used literally, they reach three-body recombination through detailed balance. Smaller artifacts stay below these bounds: values at the lowest temperatures that fall towards the next ones faster than any rate of the kind, such as a recombination coefficient of $2.6 \times 10^{-8}$ followed by $10^{-30}\,\mathrm{cm}^3\,\mathrm{s}^{-1}$ in OpenADAS `cop98#10_ls#ne1`. Recombination falls at most as $T^{-3/2}$ (radiative recombination into levels bound by less than $kT$, dielectronic recombination above its resonances), and reduced ionization rises as $T^{1/2}$ near threshold and falls as about $T^{-1/2}$ far above it. Points at the lowest temperatures that fall faster than $T^{-2}$ (recombination) or $T^{-1}$ (reduced ionization) towards a later point are left out with a warning, from the lowest up to the first that does not.

Radiative and dielectronic recombination supplied by a file are collected in $\alpha_{ip}^{\mathrm{RR+DR}}$. The effective recombination coefficient used at a particular density is

$$R_{ip}(T_\mathrm{e}, n_\mathrm{e}) = \alpha_{ip}^{\mathrm{RR+DR}}(T_\mathrm{e}) + n_\mathrm{e}\alpha_{ip}^{(3)}(T_\mathrm{e}). \tag{3.5}$$

Its contribution to a matrix element is $n_\mathrm{e}R_{ip}$, so the three-body term enters with two factors of electron density. Zero or missing file data must be distinguished from a reliable physical assertion that a process is absent.

ECIP is an optional approximate completion of missing ionization channels. ecip_rates prepares the allowed parent channels, the levels marked $\{p\}$ in the level list that lie below the ionization potential of parent $p$ (also above the first ionization potential), and passes Kelvin temperatures to r8ecip. suppliment_with_ecip chooses channels using parent annotations and zero entries in the assembled ionization array. This is an approximation to missing atomic data, not a substitute for a convergence study or a higher-quality calculated rate.

## 3.4 Three-body recombination and Saha detailed balance

Equilibrium of $i + e \rightleftharpoons p + e + e$ requires

$$N_i^* n_\mathrm{e} S_{ip} = N_p^* n_\mathrm{e}^2 \alpha_{ip}^{(3)}. \tag{3.6}$$

Using the electron translational partition factor gives

$$\frac{\alpha_{ip}^{(3)}}{S_{ip}} = \frac{g_i}{g_p}\frac{h^3}{2(2\pi m_e k_B T_\mathrm{K})^{3/2}}\exp\left(\frac{\chi_{ip}}{T_\mathrm{e}}\right) \simeq \frac{g_i}{g_p}\frac{1.6564\times10^{-22}}{T_\mathrm{e}^{3/2}}\exp\left(\frac{\chi_{ip}}{T_\mathrm{e}}\right)\ \mathrm{cm}^3. \tag{3.7}$$

The last numerical form takes $T_\mathrm{e}$ in eV and includes the free-electron spin factor of two. The implementation uses the close approximation $1.656742\times10^{-22}$.

**Stable low-temperature evaluation.** Three-body recombination uses the reduced ionization coefficient directly, avoiding a product of an underflowed ionization rate and an overflowing inverse exponential. This gives a finite positive result at a fixed positive temperature when the reduced coefficient is finite and positive; it does not establish a universal $T \to 0$ limit.

**Standalone LTE helpers.** boltzmann_population returns ground-relative level populations or total-normalized fractions. saha_population includes both stage partition functions, ionization energy, electron mass, and electron density; it returns $N_{z+1}/N_z$ or the two-stage ionized fraction. Energies are in $\mathrm{cm}^{-1}$, temperature in eV, and electron density in $\mathrm{cm}^{-3}$.

### 3.5 Thermal charge exchange

A donor species at temperature $T_h$ and density $n_\mathrm{h}$ can capture an electron into bound level $i$ from parent $p$. Writing its coefficient $K_{ip}(T_h)$, the gain is $n_\mathrm{h}K_{ip}N_p^\mathrm{CX}$. The density multiplier is the donor density, not automatically the electron density. The routine make_cx_rates_from_file interpolates the donor-temperature grid. The user-supplied donor and electron grids must be compatible with the selected grid or paired layout.

## 4 Interpolation and asymptotic continuation

The interpolation option parser separates a leading `log` or `loglog` flag from the SciPy interpolation kind. The `log` flag selects *logarithmic temperature coordinates*; it does not take the logarithm of the rate coefficient. For example,

$$\widehat{r}(T_\mathrm{e}) = \mathcal{I}[\ln T_k, r_k]\,(\ln T_\mathrm{e}) \tag{4.1}$$

still interpolates the ordinates $r_k$ directly. Log-temperature interpolation alone therefore does not guarantee positive extrapolated rates: the extrapolation is clipped at zero, which the Be I recombination and ionization coefficients reached a few times above the file temperatures.

By default (`loglog_slinear`), recombination and reduced ionization coefficients are interpolated as

$$\widehat{r}(T_\mathrm{e}) = \exp\mathcal{I}[\ln T_k, \ln r_k]\,(\ln T_\mathrm{e}), \tag{4.2}$$

over the positive values of each transition. Between two temperatures this gives the geometric mean of their values where linear interpolation of $r_k$ gives the arithmetic mean, which is never smaller and overestimates coefficients that change by orders of magnitude between neighbouring temperatures (by up to 45 times for OpenADAS $\mathrm{Ne}^{7+}$ and 236 times for Be I recombination). Outside the file temperatures the coefficients follow a power law through the two nearest values, $\widehat{r} \propto T^{\,s}$, with the exponent limited to physical values: $-3/2 \le s \le 0$ for recombination, so it neither grows nor falls faster than radiative and dielectronic recombination; for reduced ionization

$-1 \leq s \leq 1/2$ above the file temperatures and $s \geq 0$ below them, where a falling coefficient is held constant. The options without `loglog` keep the interpolation of the coefficients themselves; excitation and charge exchange use them, and excitation is continued outside the file temperatures as described below.

The Burgess–Tully helpers compress an unbounded temperature coordinate into $0 \leq x < 1$. With $\theta = k_B T/\Delta E$ and a positive scaling parameter $c$ (default 1.5), the four implemented transforms are An available infinite-energy point constrains the transformed high-temperature continuation.

| Type | $x(\theta)$ | Transformed ordinate $y$ |
|---|---|---|
| 1 | $1 - \ln c/\ln(\theta + c)$ | $\Upsilon/\ln(\theta + \mathrm{e})$ |
| 2 | $\theta/(\theta + c)$ | $\Upsilon$ |
| 3 | $\theta/(\theta + c)$ | $(1 + \theta)\Upsilon$ |
| 4 | $1 - \ln c/\ln(\theta + c)$ | $\Upsilon/\ln(\theta + c)$ |

**Table 2:** Coordinate and ordinate definitions in `burgess_tully_rates.py`. The original routines use Kelvin and wavenumbers to form $\theta$.

Otherwise the code fits the final calculated points in transformed space. The inverse transform is necessary to recover $\Upsilon$.

**Continuation in development.** The caller applies all four high-temperature categories and uses one aligned fallback slot per category. Below the file grid it applies a nonnegative transformed continuation or retains the first collision strength for transitions whose data start far above threshold. This numerical policy prevents negative rates but does not establish atomic-data accuracy outside the calculated range.

# 5 The collisional radiative matrix

For bound levels $i \neq j$, define $q_{j\to i}$ as the corresponding electron-impact coefficient. Then

$$C_{ij} = A_{j\to i} + n_\mathrm{e} q_{j\to i}, \tag{5.1}$$

$$C_{ii} = -\sum_{j\neq i}(A_{i\to j} + n_\mathrm{e} q_{i\to j}) - n_\mathrm{e}\sum_p S_{ip}. \tag{5.2}$$

Here forbidden or upward spontaneous transitions have $A = 0$. For adjacent-charge parents,

$$C_{pi} = n_\mathrm{e} S_{ip}, \qquad C_{ip} = n_\mathrm{e} R_{ip}. \tag{5.3}$$

The corresponding parent diagonal loses recombination into the lower stage. A closed collection of states satisfies

$$\sum_i C_{ij} = 0 \quad \text{for every source column } j, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\sum_i N_i = 0. \tag{5.4}$$

Column sums need not vanish in a deliberately open subsystem; the missing population must then be accounted for as an external source or sink.

**Implementation.** populate_cr_matrix stores the main generator in `cr_matrix['cr']`; `A_ji` is stored with upper-level index first, so radiative entries are transposed when inserted into the generator. `cr_red` later contains only the selected excited-state block. The separate `cr_loss` array holds only the level-loss diagonal of that generator.

The CX driving columns used by the quasistatic solution have no matching losses: they attribute emission to CX from a given parent population. solve_time_dependent instead moves the CX

gains to the recombining-parent columns and subtracts them from the parent diagonal, so CX depletes the evolving parent and the transient generator satisfies Eq. (5.4). The CX columns are then empty; their initial populations and sources must be zero.

The off-diagonal sign, column sums, and inverse-channel density powers are useful checks before any matrix is solved. A linear algebra routine can solve a matrix with negative or incorrectly scaled atomic rates without identifying the underlying physical error.

# 6 Projection of the high $n$-shells

The levels of an ADF04 file usually cover the lowest principal shells, $n \le n_1$. The levels of higher shells, up to the series limits of the parents, are missing from the matrix of Section 5, yet most recombination passes through them: radiative capture into high $n$, dielectronic capture into Rydberg states, and the cascades that follow. At high density they also open stepwise ionization. For neon, the effective recombination into $Ne^{3+}$, $Ne^{4+}$ and $Ne^{5+}$ from the ADAS `cop98#10` ADF04 files alone is 0.17–0.69 of the ADAS coefficients made from the same files with the high shells included (5–100 eV), and the SCD of $Ne^{2+}$ at 1 eV and $10^{14}\,\mathrm{cm}^{-3}$ is 0.007 of the ADAS one.

## 6.1 Condensed bundle-$nS$ couplings

A bundle-$nS$ calculation (ADAS204) groups the levels built on a parent $\rho$ of the next ion with spin system $S$ into $n$-shell populations $x_n$, from the lowest shell to the series limit, and eliminates all but its lowest shells. The condensed couplings are given for each parent and spin system on a temperature–density grid (ADF17 files), in the convention of losses positive:

$$\sum_m P_{nm}\, x_m = n_\mathrm{e} N_\rho\, r_n, \tag{6.1}$$

where $P_{nm}$ ($\mathrm{s}^{-1}$) is the coupling from shell $m$ to shell $n$ ($P_{nn}$ the total loss of shell $n$), $r_n$ ($\mathrm{cm}^3\,\mathrm{s}^{-1}$) the recombination into shell $n$ by all routes, $I_{n\rho'}$ ($\mathrm{s}^{-1}$) the ionization of shell $n$ to parent $\rho'$, and $Q_{\rho'}$ ($\mathrm{cm}^3\,\mathrm{s}^{-1}$) the recombination from $\rho$ that ionizes to another parent $\rho'$ without reaching the lowest shells. The same files give the direct parts: the shell couplings $D_{nm}$, direct ionization $D^\mathrm{I}_{n\rho'}$ and direct radiative, dielectronic, three-body and charge-exchange recombination $d_n$.

Shells $u$ above the highest shell of the ADF04 levels are eliminated from Eq. (6.1) onto the kept shells $l$ (a Schur complement):

$$P' = P_{ll} - P_{lu}P_{uu}^{-1}P_{ul}, \qquad r' = r_l - P_{lu}P_{uu}^{-1}r_u, \tag{6.2}$$

$$I' = I_l - \left(P_{uu}^{-1}P_{ul}\right)^\mathsf{T} I_u, \qquad Q' = Q + I_u^\mathsf{T} P_{uu}^{-1} r_u, \tag{6.3}$$

and the direct transfers from kept to eliminated shells become indirect. The ADF04 file already has the direct couplings between its levels, so they are subtracted. The indirect couplings through the high shells are then

$$K = P' - D' - \mathrm{diag}\Big(\sum_{\rho'} D^\mathrm{I}\Big), \quad I^\mathrm{ind} = I' - D^\mathrm{I}, \quad r^\mathrm{ind} = r' - d, \quad Q^\mathrm{ind} = Q'. \tag{6.4}$$

Negative values up to $10^{-3}$ of the projected and direct values they are the difference of are round-off (the ADF17 files have 11 digits) and are set to zero. Larger ones show condensed data that are not reliable, as at the lowest temperatures of most ADF17 files; those temperatures are left out of that coupling of the parent and spin system.

## 6.2 Expansion over the levels

An ADF18 expansion file (`a17_p208` kind) assigns each ADF04 level $i$ to a parent, spin system and shell $n(i)$, with a weight $w_i$ normalised over the levels of that shell (statistical weights). Level populations add up to shell populations, so a rate per unit shell population applies to each of its levels, while a flow into a shell is shared by its levels with their weights:

$$R^{\mathrm{ind}}_{i\rho} = w_i\, r^{\mathrm{ind}}_{n(i)}, \qquad S^{\mathrm{ind}}_{i\rho} = I^{\mathrm{ind}}_{n(i)\rho}/n_{\mathrm{e}}, \tag{6.5}$$

$$G_{ij} = -w_i\, K_{n(i)n(j)} \quad (n(i) \neq n(j)), \qquad X^{\mathrm{ind}}_{\rho'\rho} = \sum_S Q^{\mathrm{ind}}_{\rho'}. \tag{6.6}$$

Transfers within a shell are not described by shell populations and are omitted. These terms are added to the matrix of Section 5: $G_{ij}$ to $C_{ij}$, $n_{\mathrm{e}}S^{\mathrm{ind}}_{i\rho}$ to $C_{\rho i}$ and $n_{\mathrm{e}}R^{\mathrm{ind}}_{i\rho}$ to $C_{i\rho}$, each with its loss on the diagonal of the source, so Eq. (5.4) still holds. The indirect parent coupling $X^{\mathrm{ind}}$ is added to XCD (Section 9.4). Populations, PECs, SCD, ACD, QCD and XCD then include the high shells.

**Choices of the implementation.** Ionization through the shells of one parent to another parent is the autoionization of core-excited Rydberg states, part of excitation-autoionization, which the ionization rates of the ADF04 files (ADF07 data) already include. It is left out. ADAS208 also credits only the group's own parent but keeps that loss on the diagonal; ColRadPy omits both, so the matrix conserves particles. The diagonal follows from the expanded gains and ionization rather than from $K_{nn}$; the two agree when the condensed data conserve particles. The condensed data are interpolated linearly in $\ln T_{\mathrm{e}}$, $\ln n_{\mathrm{e}}$ and the logarithm of each coupling, without the temperatures where it is not reliable, and held at the edge of the ADF17 grid outside it; both give a warning when the user temperatures need them. The fractional-parentage corrections of the lowest shell (switches `IEFPRE`, `IEFPRS`, set only by the superseded `cbnm93` files) are not implemented, and indirect charge-exchange recombination is not represented.

**Routines and data.** `read_adf17` reads the condensed matrices and `read_adf18` the expansion files; `projection.project` forms and expands the indirect couplings. `colradpy(..., projection=(adf17, adf18))` stores them in `rates['projection']` (`make_projection`) and `populate_cr_matrix` adds them to the generator. In an ionization balance, `projection` takes one pair, or `None`, per ADF04 file. The ADF18 file names the ADF04 and ADF17 files it maps; its level numbers follow the order of the ADF04 file, before any sorting by energy.

## 6.3 Comparison with ADAS for neon

With the ADF04 files, condensed matrices (`cbnm96`) and expansion files (`exp96`) from which ADAS made its neon ADF11 coefficients, the metastable-resolved SCD and ACD from the ground metastable agree with `scd96r` and `acd96r` to within a few per cent for most stages where they are abundant. For example, the ACD into $Ne^{3+}$, $Ne^{4+}$ and $Ne^{5+}$, from the ADF04 files of these ions, is 0.97–1.02 of ADAS from 5 to 100 eV, and their SCD 0.88–1.02 from 10 eV, where without the projection that ACD was 0.17–0.69 of ADAS. The steady fractional abundances then agree with the balance of the ADAS coefficients to within 0.06 at $10^{12}\,\mathrm{cm}^{-3}$, 0.07 at $10^{13}\,\mathrm{cm}^{-3}$ and 0.16 at $10^{14}\,\mathrm{cm}^{-3}$, against 0.34–0.83 without the projection. The ACD into $Ne^{+}$ remains 1.2–1.4 times that of ADAS, and that into $Ne^{0}$ at $10^{14}\,\mathrm{cm}^{-3}$ 0.65–0.97 of it. ADAS notes that these two stages were recalculated with a corrected expansion file, which may differ from the one used here.

The ADF17 grids are in reduced temperature, so their first temperature rises with the charge: 0.04 eV for $Ne^{0}$, 1.1 eV for $Ne^{4+}$ and 4.3 eV for $Ne^{9+}$. Below it the projection holds the edge values, with a warning, and the coefficients of the highly charged ions there are not meaningful: the indirect ionization held at the edge gives an SCD orders of magnitude above that of ADAS. At

the lowest grid temperatures the condensed data are not reliable either: `cbnm96#h_ne9ls` gives recombination of $+1.1 \times 10^9$ and $-1.1 \times 10^9\,\mathrm{cm^3\,s^{-1}}$ into $n = 1$ and $n = 2$ at 4.3 eV. Negative indirect values beyond round-off occur in the recombination of 125 of the 146 parents and spin systems of the ADAS `a17_p208` mappings, and in the parent coupling of 3, always at the lowest one to three temperatures. Without those temperatures the ACD into $Ne^{9+}$ is 0.70 of ADAS at 5 eV and 0.90 at 10 eV ($10^{12}\,\mathrm{cm^{-3}}$); keeping their positive values would give $10^{15}$ and 12.6 times ADAS. These ions are absent at such temperatures, so the balance is not affected. The comparison is not part of the test suite: the ADAS files are not distributed with ColRadPy.

# 7 Quasistatic elimination of excited levels

Partition the population vector into retained states $N_r$ and excited states $N_e$. The full system is

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} N_r \\ N_e \end{pmatrix} = \begin{pmatrix} C_{rr} & C_{re} \\ C_{er} & C_{ee} \end{pmatrix}\begin{pmatrix} N_r \\ N_e \end{pmatrix}. \tag{7.1}$$

When excited-state relaxation is fast compared with evolution of the drivers,

$$0 = C_{ee}N_e + C_{er}N_r, \qquad N_e = -C_{ee}^{-1}C_{er}N_r. \tag{7.2}$$

This requires an invertible, relaxing excited block. Including a long-lived state in the eliminated set can invalidate the approximation even if the numerical inverse exists. Time-scale separation should be assessed from the excited-block decay rates and the time scale of the retained dynamics.

Substitution into the retained equations gives the Schur complement

$$C_{\mathrm{eff}} = C_{rr} - C_{re}C_{ee}^{-1}C_{er}. \tag{7.3}$$

This formula is the common origin of the effective coefficients below. For a closed model, correctly eliminating excited states preserves column-sum conservation in the retained equations.

## 7.1 Choosing retained levels and testing time-scale separation

The paper emphasizes that a spectroscopically metastable level and a level that must be retained dynamically are not identical concepts [1]. A weak spontaneous decay can produce a large population, but collisions and ionization can shorten its actual relaxation time. Conversely, a collection of ordinary excited levels can participate in a slow, strongly coupled mode. The retained set is therefore a modeling choice for the density, temperature, initial condition and diagnostic being considered.

For constant matrix blocks, write the instantaneous quasistatic response as $N_e^{\mathrm{qs}}(t) = -C_{ee}^{-1}C_{er}N_r(t)$ and let $\delta = N_e - N_e^{\mathrm{qs}}$. The exact lag equation is

$$\dot{\delta} = C_{ee}\delta - \dot{N}_e^{\mathrm{qs}}, \tag{7.4}$$

$$\delta(t) = \mathrm{e}^{C_{ee}t}\delta(0) - \int_0^t \mathrm{e}^{C_{ee}(t-u)}\dot{N}_e^{\mathrm{qs}}(u)\,\mathrm{d}u. \tag{7.5}$$

There are two distinct errors: an initial excited-state relaxation transient and the lag behind a changing driver. After the first decays, a small ratio of excited relaxation time to driver-evolution time supports the approximation. The decaying eigenvalues and eigenvectors of $C_{ee}$ help assess this, although a nonnormal matrix can amplify transients and a gap in eigenvalues alone is not a complete error bound. Changing plasma conditions also changes the response matrix itself and contributes to $\dot{N}_e^{\mathrm{qs}}$.

A useful practical test is to start full and reduced models from consistent retained populations, vary the retained set, and compare the predicted line emission over the required observation times. A ground-only quasistatic model eliminates every other bound state, including levels that might need their own evolving populations. A large excited population identifies a candidate for examination, but a small population can still be an important feeder of a diagnostic line. Eigenmodes are combinations of levels; each eigenvalue should not be labeled as the lifetime of one level without examining its eigenvector.

## 7.2 Separate excitation, recombination and CX drivers

The excited-state forcing can be separated as

$$0 = C_{ee}N_e + C_{em}N_m + n_{\mathrm{e}}R_{ep}N_p + n_{\mathrm{h}}K_{ep}N_p^{\mathrm{CX}}. \tag{7.6}$$

Define the dimensionless response matrices

$$F = -C_{ee}^{-1}C_{em}, \qquad G = -C_{ee}^{-1}n_{\mathrm{e}}R_{ep}, \qquad H = -C_{ee}^{-1}n_{\mathrm{h}}K_{ep}, \tag{7.7}$$
$$N_e = FN_m + GN_p + HN_p^{\mathrm{CX}}.$$

Each column states a response to one independently prescribed driving population. These are not probabilities whose sums must all be one.

**Implementation.** solve_quasi_static constructs `beta`$= -C_{er}$ and computes `pops`$= C_{ee}^{-1}$`beta`. The excitation columns come first, followed by recombination and then CX columns when enabled. `processed['F']` denotes an excitation response without the additional $1/n_{\mathrm{e}}$ used in some ADAS definitions. Matching symbols without matching their density convention produces unit errors.

**CX-only construction.** When CX is enabled but both radiative and three-body recombination are disabled, the CX columns still follow the parent columns, which the generator keeps whether or not recombination is used.

## 7.3 Normalization is a choice of driving population

For a single ground driver, an excitation response $F_i = N_i/N_0$ can be converted to a fraction relative to the ion total:

$$\frac{N_0}{N_{\mathrm{ion}}} = \frac{1}{1+\sum_i F_i}, \qquad \frac{N_i}{N_{\mathrm{ion}}} = \frac{F_i}{1+\sum_j F_j}. \tag{7.8}$$

This transformation changes the denominator of an excitation coefficient. It is not applicable without modification to $G_i = N_i/N_p$, whose parent belongs to another charge state. The independent drivers are explicitly distinguished in GCR emissivity definitions [2].

**Normalization in development.** With one retained metastable, excitation columns are normalized by default; recombination and CX columns remain relative to their independent parents. With more than one retained metastable, excitation normalization defaults off. The `default_pop_norm` option controls this choice. For example, an excited level with $A = 1\,\mathrm{s}^{-1}$ and parent-driven injection $n_{\mathrm{e}}\alpha = 2\,\mathrm{s}^{-1}$ retains the required response $N_e/N_p = 2$.

### 7.4 Unsummed contributions and dominant feeding mechanisms

Paper Section 2.3 retains the individual terms of the excited-state solution to identify important populating channels [1]. Let $a$ label an excitation, recombination or CX driver, and write $\beta_{ja} = -C_{ja}$ for the negative of its driving-block entry into eliminated level $j$. Define

$$U_{ija} = (C_{ee}^{-1})_{ij}\beta_{ja} \quad \text{(no sum over } j\text{)}, \qquad N_i = \sum_a \sum_j U_{ija} N_a. \tag{7.9}$$

The sum over $j$ gives the response of level $i$ to driver $a$; leaving it unsummed assigns that response to the level where the driver injects population into the eliminated manifold. For a fixed driver with a nonzero total response, the fractions

$$f_{j|i,a} = \frac{U_{ija}}{\sum_k U_{ika}} \tag{7.10}$$

identify the injection channels carrying the largest contribution. This interpretation requires physical nonnegative rates and a relaxing excited block. A common normalization applied to one driver's response cancels in these fractions, but different drivers must be weighted by their actual populations before they are combined.

**The stored contribution tensor.** The implementation stores $U$ in `processed['pop_lvl']` with axes *destination excited level, injection excited level, driver, temperature, density*. Both excited-level indices refer to the reduced level map, not to physical bound-level numbers. Summing the second axis reconstructs `processed['pops']` under the same normalization. Paired mode replaces the last two axes by one pair index.

An entry with $j = i$ represents direct injection into the observed upper level, followed by its full response through $C_{ee}^{-1}$. An entry with $j \neq i$ represents injection elsewhere followed by redistribution or a cascade. These are algebraic injection-channel contributions, not exclusive microscopic paths or counts of the last collision: the inverse already contains repeated transfers. This qualification matters when interpreting the paper's direct-excitation and cascade illustrations for Mo I.

The same accounting can resolve effective rates. For example, level $i$ contributes $S_{ip}F_{i\rho}$ to indirect ionization, and contributes $C_{\rho i}G_{ip}/n_\mathrm{e}$ to capture feeding retained level $\rho$. Keeping these terms unsummed can guide a reduced atomic model, provided its emission and effective-rate errors are checked afterward.

## 8 Photon emission and power coefficients

For an optically thin line $u \to \ell$, the angle-integrated photon emissivity is

$$\epsilon_{u\ell} = A_{u\ell} N_u \quad [\text{photons}\,\text{cm}^{-3}\,\text{s}^{-1}]. \tag{8.1}$$

Using Eq. (7.7), the three types of coefficient are

$$\mathrm{PEC}^{\mathrm{exc}}_{u\ell,\rho} = A_{u\ell} F_{u\rho}/n_\mathrm{e}, \tag{8.2}$$

$$\mathrm{PEC}^{\mathrm{rec}}_{u\ell,p} = A_{u\ell} G_{up}/n_\mathrm{e}, \tag{8.3}$$

$$\mathrm{PEC}^{\mathrm{CX}}_{u\ell,p} = A_{u\ell} H_{up}/n_\mathrm{h}. \tag{8.4}$$

Consequently the emitted photon rate is

$$\epsilon_{u\ell} = n_\mathrm{e} \sum_\rho N_\rho \mathrm{PEC}^{\mathrm{exc}}_{u\ell,\rho} + n_\mathrm{e} \sum_p N_p \mathrm{PEC}^{\mathrm{rec}}_{u\ell,p} + n_\mathrm{h} \sum_p N^{\mathrm{CX}}_p \mathrm{PEC}^{\mathrm{CX}}_{u\ell,p}. \tag{8.5}$$

For an isotropic emitter, a line-of-sight photon radiance is $\int \epsilon_{u\ell}\,\mathrm{d}s/(4\pi)$. The $4\pi$ factor is an observation convention and is not already part of $A_{u\ell}N_u$.

**CX PEC convention.** The implementation divides *all* computed driving columns by $n_\mathrm{e}$ when forming `processed['pecs']`. Its stored CX columns therefore contain $AH/n_\mathrm{e} = (n_\mathrm{h}/n_\mathrm{e})\mathrm{PEC}^{\mathrm{CX}}$ as defined above. Use electron density with that stored response, or explicitly convert to a donor-normalized coefficient before using Eq. (8.5).

If $E_\gamma = hc/\lambda_\mathrm{vac}$, the radiated power coefficient associated with any PEC is

$$P[\mathrm{W\,cm^3}] = E_\gamma[\mathrm{J}]\,\mathrm{PEC}, \tag{8.6}$$

$$P[\mathrm{W\,m^3}] = E_\gamma[\mathrm{J}]\,\mathrm{PEC}\times 10^{-6}. \tag{8.7}$$

Sum excitation columns over lines for a PLT-like cooling coefficient and recombination columns for a PRB-like coefficient, retaining the driver index. A total plasma cooling rate additionally needs the corresponding densities. The stored `prb` sums recombination-driven line power; CX-driven power is recorded separately as `prc`.

**Power outputs.** Quasistatic and time-dependent line power use the physical upper-level population and the exact photon energy from the level-energy difference. Their coefficients are in $\mathrm{W\,cm^3}$ and satisfy line power = $E_\gamma$ times PEC for the same line and driver.

## 8.1 S/XB: ionizations per emitted photon

Table 1 of the paper includes the diagnostic ratio S/XB, which relates ionization to excitation-driven line emission [1]. For a single retained driver $\rho$, define the total effective ionization coefficient $\mathcal{S}_\rho = \sum_p \mathcal{S}_{\rho p}$. For line $u \to \ell$,

$$(S/XB)_{u\ell,\rho} = \frac{\mathcal{S}_\rho}{\mathrm{PEC}^{\mathrm{exc}}_{u\ell,\rho}}. \tag{8.8}$$

Both coefficients use the same electron-density and driving-population convention, so the ratio is dimensionless: ionizations per excitation photon. The numerator can instead be restricted to a specified destination parent, provided that restricted meaning is stated.

For multiple retained populations, form the rate ratio before interpreting it as a diagnostic:

$$(S/XB)_{u\ell}[N_m] = \frac{\sum_\rho N_\rho \mathcal{S}_\rho}{\sum_\rho N_\rho \mathrm{PEC}^{\mathrm{exc}}_{u\ell,\rho}}. \tag{8.9}$$

This is not an unweighted average of the individual metastable ratios. Different metastable fractions can produce different ionization-to-photon ratios at the same temperature and density.

If excitation dominates the observed line and S/XB is approximately constant through the emitting column, its photon radiance $I_{u\ell}$ gives

$$\int \dot{n}_\mathrm{ion}\,\mathrm{d}s = (S/XB)_{u\ell}\,4\pi I_{u\ell}. \tag{8.10}$$

If the ratio varies spatially, the local ionization rate must instead be integrated with the corresponding emissivity. Interpreting that column ionization rate as a surface erosion or influx rate additionally requires a geometry and transport model: escape without ionization, repeated recombination and ionization, and recombination/CX photons can invalidate a simple one-ionization-per-entering-particle interpretation. A zero excitation PEC does not define a finite ratio.

**Computing S/XB from development output.** Use a parent sum of `scd` and the matching excitation column of `pecs`, with consistent normalization. Despite its docstring, the quasistatic routine does not store a computed SXB array. The repository has a separate `read_adf13` reader for tabulated SXB data. The paper's compact coefficient notation should not be used to mix a per-driver photon rate in $s^{-1}$ with an ionization coefficient in $cm^3/s$.

# 9 Generalized collisional radiative coefficients

Effective coefficients combine direct processes with paths through eliminated excited states. Their definition is a reduction of the generator, not simply a sum of raw atomic rates. All formulas in this section assume the unnormalized responses of Eq. (7.7) and explicitly prescribed retained populations.

## 9.1 SCD: effective ionization

For a retained bound level $\rho$ and destination parent $p$,

$$\mathcal{S}_{\rho p} = S_{\rho p} + \sum_i S_{ip} F_{i\rho}. \tag{9.1}$$

The second term describes excitation or redistribution followed by ionization. Both terms have coefficient units. Its matrix contribution from $\rho$ to $p$ is $n_e \mathcal{S}_{\rho p}$.

## 9.2 ACD and CCD: effective capture

The capture coefficient from parent $p$ into retained level $\rho$ is

$$\mathcal{A}_{\rho p} = R_{\rho p} - \sum_{ij} C_{\rho i} (C_{ee}^{-1})_{ij} R_{jp}. \tag{9.2}$$

The second term incorporates capture into excited levels followed by their collisional redistribution and radiative cascades. Replacing $R$ by $K$ defines $\mathcal{C}_{\rho p}$, the CCD coefficient. The rate multipliers are $n_e$ and $n_h$, respectively. In an ionization matrix expressed per electron density, the CCD entry carries $n_h/n_e$.

## 9.3 QCD: redistribution within a retained bound manifold

For two different retained bound levels,

$$\mathcal{Q}_{\sigma\to\rho} = \frac{C_{\rho\sigma} + \sum_i C_{\rho i} F_{i\sigma}}{n_e}, \qquad \rho \neq \sigma. \tag{9.3}$$

This includes direct mixing and indirect paths through the excited manifold. The diagonal is kept zero in the stored QCD array; its population loss is inserted separately when the charge-state matrix is assembled.

## 9.4 XCD: redistribution between parent metastables

A parent $p$ can recombine into an excited level of the lower stage and then ionize into a different parent $q$. The coefficient for this path is

$$\mathcal{X}_{p\to q} = -\sum_{ij} C_{qi} (C_{ee}^{-1})_{ij} R_{jp} = -n_e \sum_{ij} S_{iq} (C_{ee}^{-1})_{ij} R_{jp}, \qquad p \neq q. \tag{9.4}$$

The factor $n_e$ in the second form is essential. Dimensional analysis gives

$$[n_e S][C_{ee}^{-1}][R] = (s^{-1})(s)(cm^3\,s^{-1}) = cm^3\,s^{-1}. \tag{9.5}$$

This is equivalent to dividing the parent-to-parent off-diagonal block of the Schur complement by electron density. The recombination rows must use the same excited-level ordering as $C_{ee}$. The explicit electron-density factor is also present in the paper's Table 1 definition of XCD [1].

**XCD in development.** The contraction uses $n_e S$ and selects recombination rows by the actual eliminated-level map, including noncontiguous metastable selections such as $[0, 2]$. The resulting coefficient has units $cm^3/s$ and agrees with the parent block of the direct Schur complement divided by electron density.

| Array | First index | Second index | Rate multiplier |
|---|---|---|---|
| `scd` | source bound metastable | destination parent | $n_e$ |
| `acd` | destination bound metastable | source parent | $n_e$ |
| `ccd` | destination bound metastable | source CX parent | $n_h$ |
| `qcd` | source bound metastable | destination bound metastable | $n_e$ |
| `xcd` | source parent | destination parent | $n_e$ |

**Table 3:** Stored coefficient orientation. Equal matrix dimensions can conceal an accidental transpose, so checking shapes alone is insufficient.

# 10 Time-dependent evolution and matrix exponentials

For fixed plasma conditions and an optional constant source $s$,

$$\dot{N} = CN + s, \qquad N(t) = e^{Ct}N(0) + \int_0^t e^{C(t-u)}s\,du. \tag{10.1}$$

If $C = V\Lambda V^{-1}$ has a complete eigenbasis, the source-free result is $Ve^{\Lambda t}V^{-1}N(0)$. Each source-driven mode obeys

$$v_k(t) = v_k(0)e^{\lambda_k t} + c_k\phi(\lambda_k, t), \quad c = V^{-1}s, \quad \phi(\lambda, t) = \begin{cases} (e^{\lambda t} - 1)/\lambda, & \lambda \neq 0, \\ t, & \lambda = 0. \end{cases} \tag{10.2}$$

For small $|\lambda t|$, evaluating the numerator with `expm1` avoids subtracting nearly equal floating-point numbers. Small $|\lambda|$ by itself does not make a mode negligible: its decay is governed by $|\lambda t|$. An alternative formulation is the augmented exponential

$$\begin{pmatrix} N(t) \\ 1 \end{pmatrix} = \exp\left[t\begin{pmatrix} C & s \\ 0 & 0 \end{pmatrix}\right]\begin{pmatrix} N(0) \\ 1 \end{pmatrix}. \tag{10.3}$$

For constant $C$, these formulas propagate the initial state at $t = 0$; the first requested output time is not a new integration origin.

## 10.1 Complex modes and real populations

A real, nonsymmetric rate matrix can have complex conjugate eigenvalues. For example, a conservative three-state cycle has

$$C = \begin{pmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad \lambda = 0,\ -\tfrac{3}{2} \pm \tfrac{\sqrt{3}}{2}\,i. \tag{10.4}$$

The complex modes are legitimate and must be retained in the reconstruction. For real initial data and a real source, their imaginary contributions cancel in the exact populations. Taking the real part of eigenvectors or eigenvalues changes the dynamics. Taking the real part of a reconstructed population is justified only after a numerical assessment of its imaginary residual. Real-arithmetic matrix-exponential methods avoid those complex intermediates.

### 10.2 Defective and ill-conditioned eigenbases

Not every valid generator is diagonalizable. An equal-rate chain is a useful counterexample:

$$C = \begin{pmatrix} -1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad N(0) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad N(t) = \begin{pmatrix} \mathrm{e}^{-t} \\ t\mathrm{e}^{-t} \\ 1 - (1+t)\mathrm{e}^{-t} \end{pmatrix}. \tag{10.5}$$

The $t\mathrm{e}^{-t}$ term comes from a Jordan block. A nominal inverse of numerical eigenvectors can return plausible values that omit this term. Near a defective matrix, eigenvector conditioning matters more than the visual separation of eigenvalues alone. A robust algorithm should assess that conditioning and use a matrix-exponential or equivalent stable method when needed.

**Transient algorithms.** The transient solvers check eigenvector conditioning and compare selected populations against a matrix exponential. Untrusted points are recomputed with scipy.linalg.expm; a constant source uses an augmented matrix. Real populations are returned, and slow physical modes are retained according to their evolution over the requested times rather than an absolute rate cutoff. The cached evaluator needs the original rate matrix to perform the same exponential fallback.

### 10.3 Stationary solutions

A normalized stationary population of a closed model satisfies

$$C\pi = 0, \qquad \mathbf{1}^{\mathsf{T}}\pi = 1, \qquad \pi_i \geq 0. \tag{10.6}$$

For a unique equilibrium, one redundant balance equation can be replaced by the normalization equation. Rescaling balance equations helps avoid loss of weak-rate constraints. If multiple disconnected closed classes exist, their relative weights depend on the initial condition; normalization alone does not determine a unique distribution.

A rate matrix that conserves particles (non-negative off-diagonal rates and columns that sum to zero) is solved by Grassmann–Taksar–Heyman elimination, which uses only the off-diagonal rates and has no subtractions, so every population keeps the relative accuracy of the rates however small it is. Solutions through the eigenvectors or a linear solve are accurate only relative to the largest population: for a resolved carbon balance at 10 eV they gave $\mathrm{C}^{5+}$ $2 \times 10^{-5}$ where the exact value is $3 \times 10^{-16}$. For other matrices the stationary solver tries the smallest-magnitude eigenmode and checks its residual and normalization; an unreliable projection is replaced by a normalized stationary solve. The ionization-balance wrapper calls this routine directly.

## 11 Ionization balance across charge stages

Collect all retained metastables across the modeled stages into a vector $x$. At fixed plasma parameters,

$$\dot{x} = Mx + s, \qquad M = n_{\mathrm{e}}B. \tag{11.1}$$

Here $B$ is a coefficient matrix with units $\mathrm{cm}^3/\mathrm{s}$. Its adjacent-stage blocks contain SCD upward and ACD downward. QCD connects retained levels within the lower stage, and XCD contributes connections within the parent stage through the lower stage. A given stage can therefore receive both its own QCD and XCD contributions computed from the preceding charge state.

For each off-diagonal transition, the same coefficient is subtracted from the source diagonal. Consequently $\mathbf{1}^{\mathsf{T}}B = 0$ in a closed calculation. The CX downward blocks use $(n_{\mathrm{h}}/n_{\mathrm{e}})\mathcal{C}$ inside $B$, so the final multiplication by electron density recovers the physical multiplier $n_{\mathrm{h}}$.

**Implementation.** populate_ion_matrix stores $B$ in `ion_matrix`, despite that name suggesting a density-weighted rate. solve_no_source and solve_source multiply by the electron-density grid before invoking the matrix-exponential solvers. Do not apply the density multiplier a second time when interpreting their populations.

### 11.1 Ionization age

For constant electron density, define $\tau = n_{\mathrm{e}}t$. If $B$ has no density dependence, a solution at fixed temperature can be written as $x(\tau) = \exp(B\tau)x(0)$. In a full GCR model, quenching and three-body terms can make $B$ density-dependent, so equal ionization age does not necessarily imply equal charge-state fractions at different densities.

**Ionization-age grids.** The wrapper constructs times as $t = \tau/n_{\mathrm{e}}$ with axes *density, time* when there are multiple densities; one density uses a one-dimensional time array. This agrees with the matrix-exponential solver's grid convention, including singleton axes.

### 11.2 Input routes and retained manifolds

ionization_balance can build coefficients from ADF04 models or read precomputed ADF11/HDF5 data. The number of retained metastables is a stage-dependent quantity. A ragged collection such as $[\,[0,1],\,[0,1,2],\,]$ must not be forced into a rectangular numeric array. The order of input files also carries charge-stage information and should be explicit in a calculation record. The current readers and constructors impose additional restrictions listed in Section 17.

**Constructor defaults.** For ADF04, a flat metastable array selects levels in the first stage; later selections use the preceding SCD parent count. A per-stage list may have unequal lengths. Default initial abundances are sized after the GCR stages are loaded, including the final parent stage, and place all population in the first state. ADF11 and HDF5 source solves use the same default-abundance helper after their ionization matrix is constructed.

## 12 Charge-state energy balance

The energy-balance extension first solves the charge-state fractions $x_z(t)$, then evolves a kinetic-energy variable per modeled particle,

$$\mathcal{E}_z = \frac{3}{2}x_z T_z. \tag{12.1}$$

It transports that variable between charge states with the same ionization matrix and adds exchange with a prescribed background ion bath at $T_b$:

$$\dot{\mathcal{E}}_z = \sum_{z'} M_{zz'}\mathcal{E}_{z'} - \frac{2}{3}\nu_z\mathcal{E}_z + \nu_z x_z T_b. \tag{12.2}$$

The convention for $\nu_z$ is an energy-transfer frequency: $\frac{3}{2}x_z\dot{T}_z = -\nu_z x_z(T_z - T_b)$ when the population is fixed. The temperature-relaxation frequency is therefore $2\nu_z/3$. This is kinetic-temperature exchange, not a complete plasma energy equation including radiation, electron heating, and ionization-potential energy losses.

### 12.1 Charged and neutral energy transfer

The charged helper implements

$$v_s^2 = 2eT_s/m_s, \tag{12.3}$$

$$\nu_{ab} = \frac{n_b Z_a^2 Z_b^2 e^4 \ln\Lambda}{\pi^{3/2}\epsilon_0^2 m_a m_b (v_a^2 + v_b^2)^{3/2}}. \tag{12.4}$$

In these expressions temperatures are numeric values in eV, $eT$ is an SI energy, masses are in kg, and $n_b$ is in $\mathrm{m}^{-3}$. The helper evaluates

$$\lambda_D^{-2} = \sum_s \frac{n_s Z_s^2 e^2}{\epsilon_0 e T_s}, \tag{12.5}$$

$$b_{\mathrm{cl}} = \frac{|Z_a Z_b| e^2}{4\pi\epsilon_0 \mu (v_a^2 + v_b^2)/2}, \qquad b_{\mathrm{q}} = \frac{\hbar}{\mu\sqrt{v_a^2 + v_b^2}}, \tag{12.6}$$

$$\ln\Lambda = \ln\frac{\lambda_D}{\max(b_{\mathrm{cl}}, b_{\mathrm{q}})}, \qquad \mu = \frac{m_a m_b}{m_a + m_b}. \tag{12.7}$$

The screened, weakly coupled collision model is not reliable when its assumptions fail or its logarithm ceases to be physically meaningful.

For the neutral, the implemented polarization-scattering approximation uses

$$K_{\mathrm{mt}} = 2.21\pi\alpha_{\mathrm{f}} c\sqrt{\frac{m_e}{\mu}\, a_{\mathrm{pol}} a_0^4}, \qquad \nu_{nb} = \frac{3 n_b \mu}{m_n + m_b} K_{\mathrm{mt}}. \tag{12.8}$$

Here $a_{\mathrm{pol}}$ is the numeric polarizability in Bohr-radius cubed and $\alpha_{\mathrm{f}}$ is the fine-structure constant. This elastic approximation does not include every long-range or inelastic energy-transfer mechanism.

**Energy extension.** The caller converts electron density from $\mathrm{cm}^{-3}$ to $\mathrm{m}^{-3}$ for the Debye helper. The energy solver uses one temperature/density point, reconstructs populations from stored eigenpairs, and integrates Eq. (12.2) with BDF. A failed integration gives a warning with the solver message, and the stored times, abundances, energies and temperatures stop at the last time reached.

## 13 Wavelengths, fine structure and line profiles

### 13.1 Wavelengths and NIST energy matching

Vacuum wavelength follows Eq. (2.2). For a chosen air convention, $\lambda_{\mathrm{air}} = \lambda_{\mathrm{vac}}/n(\lambda_{\mathrm{vac}})$. The air-wavelength helper uses a pressure- and temperature-adjusted dispersion expression:

$$D(\xi) = 8342.54 + \frac{2406147}{130 - \xi^2} + \frac{15998}{38.9 - \xi^2}, \quad \xi = 1000/\lambda_{\mathrm{vac,nm}}, \tag{13.1}$$

$$n = 1 + 10^{-8}\left[D(\xi)\frac{P}{96095.43}\frac{1 + 10^{-8}(0.601 - 0.00972T_C)P}{1 + 0.0036610T_C} - P_w(0.037345 - 0.000401\xi^2)\right]. \tag{13.2}$$

Here $P$ and water-vapor pressure $P_w$ are in Pa and $T_C$ is in Celsius; the defaults are 101325 Pa, 0 Pa and 15 °C. This empirical expression is not a wavelength-independent physical law. It has mathematical poles near 87.706 and 160.334 nm. Development retains vacuum wavelengths below 200 nm and applies an air conversion only above that threshold. An optical air convention is thus separate from the vacuum convention used for UV emission.

NIST matching changes line positions by replacing matched level energies. It does not recompute collision strengths or establish that LS splitting reproduces a resolved collisional radiative calculation. Configurations, term labels, $J$ values, and ambiguous repeated labels must be matched consistently. The development path uses the public NumPy string API and constructs split structure when needed before shifting levels.

## 13.2 Statistical splitting of LS multiplets

For a spin-allowed electric-dipole LS transition, relative component strengths can be expressed through an angular factor

$$W_{J_u J_\ell} \propto (2J_u+1)(2J_\ell+1)\begin{Bmatrix} L_u & J_u & S \\ J_\ell & L_\ell & 1 \end{Bmatrix}^2. \tag{13.3}$$

Normalize the allowed factors over the multiplet and distribute the term PEC by $\mathrm{PEC}_{J_u J_\ell} = \mathrm{PEC}_{\mathrm{LS}} W / \sum W$. This describes relative LS angular strengths with the associated statistical population assumption. It does not add resolved-$J$ collisional dynamics or configuration mixing. Development evaluates equivalent Condon–Shortley factors and renormalizes them in split_pec_multiplet for spin-conserving dipole multiplets ($|\Delta L| \leq 1$). Intercombination lines and lines with $|\Delta L| > 1$, which these factors do not describe, are kept unsplit at the wavelength between the statistically weighted term centres.

## 13.3 Doppler and natural profiles

For an emitter of mass $m$ at temperature $T_i$,

$$\Delta\nu_D = \frac{\nu_0}{c}\sqrt{\frac{2eT_i}{m}}, \qquad \phi_D(\nu) = \frac{\exp[-(\nu-\nu_0)^2/\Delta\nu_D^2]}{\sqrt{\pi}\Delta\nu_D}. \tag{13.4}$$

This is a Gaussian with standard deviation $\Delta\nu_D/\sqrt{2}$ and FWHM $2\sqrt{\ln 2}\,\Delta\nu_D$. The broadening API takes $T_i$ in eV and the emitter mass in kg.

For lifetime broadening of an upper and lower state,

$$\Delta\nu_L = \frac{\Gamma_u + \Gamma_\ell}{2\pi}, \qquad \gamma = \frac{\Delta\nu_L}{2}, \qquad \phi_L(\nu) = \frac{\gamma/\pi}{(\nu-\nu_0)^2+\gamma^2}. \tag{13.5}$$

The Lorentzian FWHM is $\Delta\nu_L$. Density-dependent pressure broadening requires additional physics; it is not supplied merely by this lifetime width.

For a profile tabulated per wavelength rather than per frequency, preserve area with $\phi_\lambda = \phi_\nu|\mathrm{d}\nu/\mathrm{d}\lambda|$ and consistent length units. The code evaluates a frequency-shaped kernel on a wavelength grid and renormalizes its finite-grid area numerically. That construction is a useful narrow-line profile but should not be assumed identical to an exact Jacobian-transformed wide profile. Grid width and resolution control omitted wings and numerical normalization.

**Profile grid resolution.** Natural profiles are evaluated on offsets $\Delta\lambda$ from the line centre, stored in `dwave_arr`, with the frequency offsets $\Delta\nu = -c\,\Delta\lambda/[\lambda_0(\lambda_0 + \Delta\lambda)]$. A natural width can be far below the float64 resolution of the wavelength itself ($5 \times 10^{-15}$ nm for the Be I line at 454.8 nm, where float64 resolves $6 \times 10^{-14}$ nm), and frequencies near $10^{15}$ Hz are resolved only to 0.1 Hz, so neither absolute grid can carry such profiles. The absolute wavelengths `wave_arr` of these lines coincide.

# 14 Relation to the published ColRadPy study

## 14.1 The reported ADAS comparison

The original paper reports a comparison with ADAS for $C^{2+}$, chosen because metastables are present in both the modeled ion and its $C^{3+}$ parents [1]. The authors report agreement of level populations and GCR coefficients within numerical rounding over their tested parameter ranges. That is a published result for the calculation described in the paper. It is not a new benchmark of the development implementation, and the manuscript does not provide the full inputs and grids needed here to reproduce every reported comparison. The current limitations in this manual remain relevant despite that historical comparison.

## 14.2 Be I and Mo I: choosing the dynamical manifold

The paper's full time-dependent model evolves every included level, whereas its conventional comparison retains selected ground/metastable populations and eliminates the other levels quasistatically. Its case studies illustrate why the adequacy of a retained set depends on the species and conditions:

| Published case | Plasma conditions | Reported observation |
|---|---|---|
| Be I, paper Fig. 3 | $T_e = 20$ eV, $n_e = 10^{13}$ cm$^{-3}$ | Retaining the important metastable substantially improves the time-dependent line-emission prediction over a ground-only reduction. |
| Mo I, paper Fig. 4 | $T_e = 8$ eV, $n_e = 5 \times 10^{11}$ cm$^{-3}$ | Even the paper's ten-metastable reduction does not reproduce the full time-dependent line history. |

**Table 4:** Results reported by Johnson et al. [1], not rerun for this manual. They do not establish universal numbers of retained levels for either species.

The Be example exhibits a separation between slowly relaxing modes and the faster excited manifold. The Mo example has a more gradual spectrum of relaxation rates, making a small retained set less successful. This motivates the explicit lag and emission-convergence tests of Section 7.1. A ground-only reduction also loses the ability to represent an independently evolving metastable fraction.

Paper Figures 2 and 5 examine Mo I emission near 379.8 and 550.6 nm through unsummed feeding terms and population-weighted metastable contributions. The useful diagnostic distinction is between strong direct ground injection and substantial feeding through other populated levels; an unweighted PEC column alone does not give either line's contribution to the measured intensity. The required weighting is Eq. (8.5).

The study assumes initially ground-state atoms in its time-dependent application and discusses how emission can evolve as metastable populations build up after a particle leaves a surface.

Those initial conditions are a physical assumption about the injected species. ColRadPy's rate equations accept a general initial vector; the time required to form metastables should not automatically be equated with a distance from a surface without particle motion and geometry.

### 14.3 Extrapolation and possible model reduction

The Mo I collision calculation used in the paper ends near 17 eV, motivating its high-temperature Burgess–Tully illustration. This is a limit of that dataset, not a temperature limit of ColRadPy. The infinite-energy information constrains a continuation but does not validate an arbitrary extrapolation far from the calculated range. The current call-path limitations are described in Section 4.

The paper also discusses bundling as a possible way to reduce the dynamical size of complex heavy-species systems. Choosing a short list of retained metastables is not by itself a population-preserving bundling algorithm. No automatic bundled model or reproduction of those proposed future calculations is supplied by this manual.

**Paper-to-manual coverage.** Paper Section 2.1 corresponds to Sections 5–8; Section 2.2 and Table 1 to Sections 8.1 and 9; Section 2.3 to Section 7.4; Section 3.1 to Section 4; and the retained-state comparisons in Section 4.1 to Section 7.1 and this section. The energy extension, profiles, and implementation caveats in this manual cover additional source-code behavior beyond that six-page article.

## 15 Analytic examples and development checks

### 15.1 A two-level density scan

For a two-level ion with upward and downward coefficients $q_{01}, q_{10}$ and radiative rate $A$, the stationary, ion-total-normalized excited fraction is

$$f_1 = \frac{n_e q_{01}}{A + n_e(q_{01} + q_{10})}, \qquad \mathrm{PEC}_{10} = \frac{A q_{01}}{A + n_e(q_{01} + q_{10})}. \tag{15.1}$$

At low density, PEC approaches $q_{01}$ and $f_1$ is proportional to density. At high density, $f_1 \to q_{01}/(q_{01}+q_{10})$ while PEC falls inversely with density. The photon rate per emitting ion approaches $Af_1$, not zero. This distinction explains why a falling PEC is not itself vanishing emission.

### 15.2 A conservative chain and a slow mode

The generator in Eq. (10.5) is closed and has a known exact solution. An independent SciPy matrix exponential reproduces it; direct execution of the development solver agrees with both it and the analytic solution while conserving population.

A second generator, $C = [[-r, 0], [r, 0]]$ with $r = 10^{-12}\,\mathrm{s}^{-1}$, has $N_0(10^{12}\,\mathrm{s}) = \mathrm{e}^{-1}$ for $N(0) = (1, 0)$. The development source solver with a zero source reproduces this slow evolution and agrees with source-free dynamics.

### 15.3 XCD checked by direct elimination

The bundled Be II file is solved with metastables $[0, 1]$, temperatures $[5, 20]$ eV and densities $[10^{12}, 10^{14}]$ cm$^{-3}$, with both recombination mechanisms and ionization enabled. At the first grid

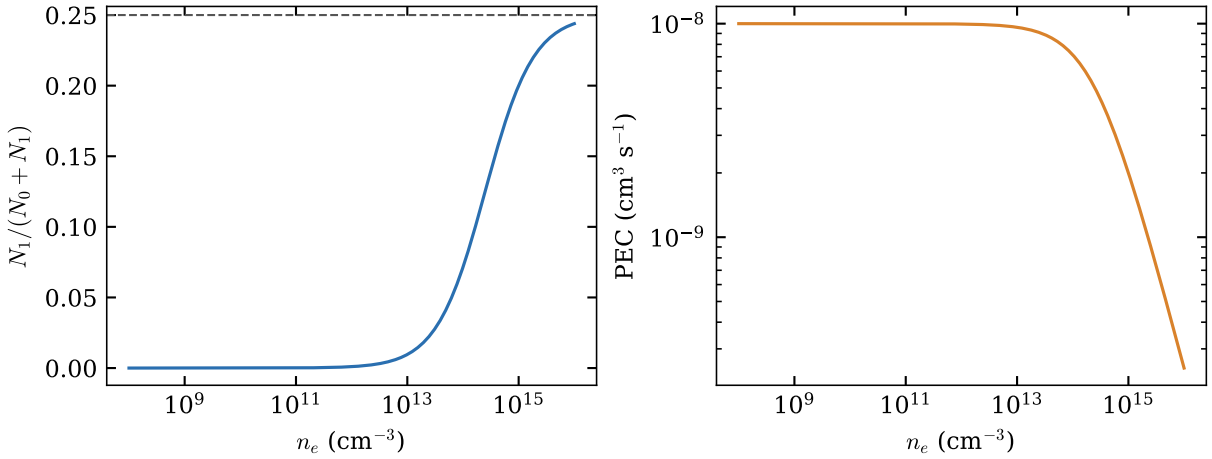



**Figure 2:** An analytic illustration with $A = 10^7\,\mathrm{s}^{-1}$, $q_{01} = 10^{-8}$ and $q_{10} = 3 \times 10^{-8}$ cm$^3$/s. The dashed line is the high-density fraction 1/4. These are illustrative coefficients, not a fit to a specific atom or an execution of the ColRadPy atomic-data path.

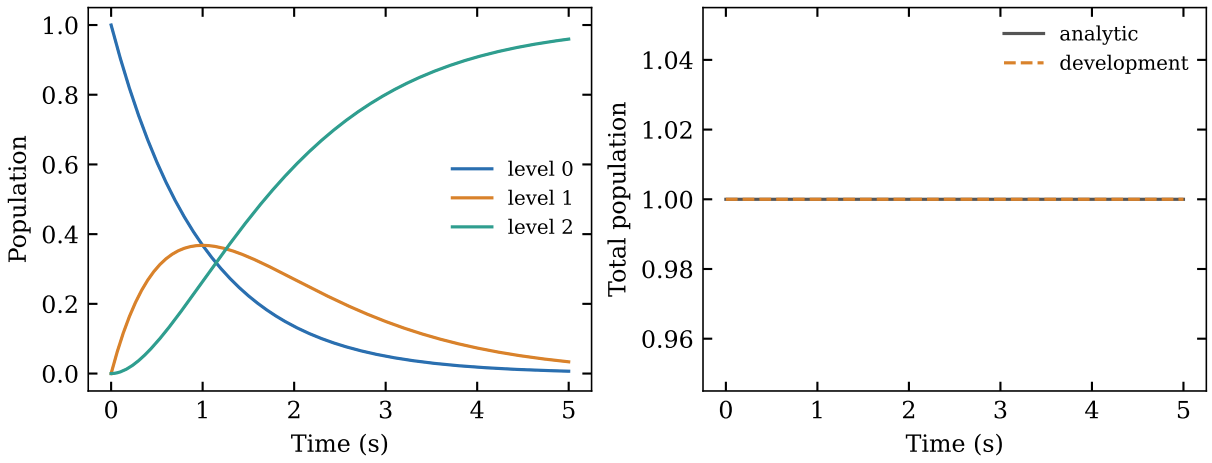



**Figure 3:** Equal-rate chain: solid curves are analytic populations; dashed curves are development results. The right panel tests the conserved sum. Agreement includes the $te^{-t}$ contribution from the defective basis.

point, form the parent block of Eq. (7.3) using a linear solve rather than the stored inverse. Divide its off-diagonal rate by $n_e$ and compare with stored XCD$[1, 0]$. Both quantities are in cm$^3$/s; a separate noncontiguous-metastable test covers row selection.

| Quantity | Development | Reference |
|---|---|---|
| Chain: $N_1(1)$ | 0.367879 | 0.367879 |
| Slow mode: $N_0(10^{12})$ | 0.367879 | 0.367879 |
| Be II: XCD$[1, 0]$ | $3.8811 \times 10^{-12}$ | $3.8811 \times 10^{-12}$ |

**Table 5:** Executed development checks. The first two rows are dimensionless populations. Both XCD values have units cm$^3$/s.

The generator records the development branch HEAD, working-tree status, package-module hashes, Python and library versions in `provenance.json`. Grid parameters and numerical results are recorded in `checks.json`. The analytic figures and checks establish specific limits and agreements. They are not a certification of all atomic datasets, an ADAS benchmark, or a reproduction of a particular tungsten dataset.

# 16 Data products and external interfaces

## 16.1 Array layout

For an ordinary temperature–density grid, thermodynamic axes trail the physical indices: $(\ldots, N_T, N_n)$. Paired mode uses $(\ldots, N_{\text{pair}})$, evaluating corresponding temperature and density entries rather than their Cartesian product. Typical arrays are

$$\begin{aligned}
\texttt{pops}: &\ (N_e, N_{\text{driver}}, N_T, N_n),\\
\texttt{pecs}: &\ (N_{\text{line}}, N_{\text{driver}}, N_T, N_n),\\
\texttt{scd}, \texttt{acd}: &\ (N_m, N_p, N_T, N_n),\\
\texttt{qcd}: &\ (N_m, N_m, N_T, N_n),\\
\texttt{xcd}: &\ (N_p, N_p, N_T, N_n).
\end{aligned}$$

The paired layout removes one grid axis. An excited-level axis is not a physical bound-level index; use the stored level map. Likewise the driving column index combines several process types and does not universally mean an ionization parent.

## 16.2 ADF04, ADF11, ADF13, ADF15 and HDF5

ADF04 is a level-resolved atomic-data format, whereas ADF11 holds effective charge-state coefficients on temperature and density grids. The ADF11 reader stores logarithmic data and the ionization-balance path interpolates the logged grids and coefficients before converting to physical values. File naming and block interpretation identify coefficient families. The separate ADF13 reader loads tabulated SXB ratios; it does not add a new SXB computation to the quasistatic solution.

ADF15 records line photon emission coefficients with wavelength and driver identifiers. Its wavelength convention is Angstroms [4], so a conversion from internal nm must occur exactly once. The number of output blocks must include every emitted line–driver pair; each block needs an unambiguous index and process/driver metadata. These are part of the scientific interface because a downstream reader can silently drop or misidentify lines.

**Export and reading.** The ADF15 wrapper converts nm to Angstroms once; the writer declares every line–driver block and assigns distinct selectors, and writes the ion field (`C + 2`) and a source member name, without which the ADAS reader stops. The ADF11 constructor collects coefficient files before filling missing QCD/XCD arrays, so their order does not require SCD first. The readers follow the fixed fields of the ADAS specifications: ADF04 values are 8-character fields, so a negative value can follow the previous one without a space; ADF11 blocks are placed by their labels (IPRT, IGRD, JPRT, JGRD), whose order and meaning change with the class, including metastable resolved power files; each ADF13 block has its own grids and metastable. They were checked against the ADAS data collections: every ADF11 and ADF13 file, the condensed projection ADF17 files and a sample of ADF04 files from every collection.

The HDF5 dumping method uses an optional serialization dependency and can select PECs, GCR coefficients, and additional arrays. Record the data version, source file and hash, selected metastables, units, grid ordering, normalization convention, and any rate supplementation with each saved result.

# 17 Validity boundaries and implementation checklist

## 17.1 Physical assumptions

The model assumes a specified Maxwellian electron bath, rates appropriate to the supplied atomic resolution, and escaping photons. The quasistatic approximation additionally requires fast excited-state relaxation. Statistical LS multiplet splitting assumes the relevant coupling and population pattern. ECIP approximates missing ionization; extrapolation does not create new atomic accuracy. The energy extension describes trace species kinetics in a prescribed background. These assumptions are separate from implementation defects.

## 17.2 Development limitations

| Area | Current qualification | Discussed in |
|---|---|---|
| Projection | The fractional-parentage switches of the `cbnm93` ADF17 files and indirect charge exchange are not implemented; the ACD into $Ne^+$ differs from ADAS by 20–40 %; below the first ADF17 temperature the couplings are held at their edge values, which makes the ionization of highly charged ions far too large there. | Section 6 |
| Coupling assumptions | Intercombination and $\|\Delta L\| > 1$ lines are kept unsplit rather than split with a coupling model. | Section 13 |

The table is a guide to interpreting this development implementation. Before using a result, check coefficient signs, state and parent maps, column-sum conservation, population normalization, line-energy consistency, density powers, and the exported units. For a transient solve, compare against an independent matrix exponential at representative grid points and verify the initial condition. For a quasistatic solve, substitute the calculated excited populations back into the balance equation. Agreement with these identities does not establish the accuracy of the input atomic calculations, but a failure identifies a problem before comparison with an experiment.

# A Source-code map for development

The map below is generated from the same development working tree as the numerical checks. Paths are relative to the repository root.

| File | Entry point | Line |
|---|---|---|
| `colradpy/colradpy_class.py` | populate_data | 640 |
| same | make_ioniz_from_reduced_ionizrates | 815 |
| same | suppliment_with_ecip | 892 |
| same | make_cx_rates_from_file | 955 |
| same | make_recombination_rates_from_file | 998 |
| same | make_three_body_recombination | 1072 |
| same | make_electron_excitation_rates | 1114 |
| same | populate_cr_matrix | 1270 |
| same | solve_quasi_static | 1584 |
| same | solve_time_dependent | 2023 |
| same | split_pec_multiplet | 2123 |
| same | shift_j_res_energy_to_nist | 2540 |
| same | solve_cr | 2676 |
| same | make_nat_broad | 2716 |
| same | make_dopp_broad | 2794 |
| same | write_pecs_adf15 | 3120 |
| `colradpy/ionization_balance_class.py` | populate_ion_matrix | 435 |
| same | solve_no_source | 583 |
| same | solve_source | 621 |
| same | solve_time_independent | 686 |
| `colradpy/solve_matrix_exponential.py` | solve_matrix_exponential | 245 |
| same | eval_matrix_exponential_solution | 318 |
| same | solve_matrix_exponential_steady_state | 349 |
| same | solve_matrix_exponential_source | 442 |
| `colradpy/energy_balance.py` | solve | 88 |
| same | plasma_energy_transfer_frequency | 243 |
| same | neutral_energy_transfer_frequency | 298 |
| same | coulomb_logarithm | 351 |
| same | debye_length | 393 |

# References

[1] C. A. Johnson, S. D. Loch and D. A. Ennis, "ColRadPy: A Python collisional radiative solver," *Nuclear Materials and Energy* (2019). doi:10.1016/j.nme.2019.01.013. The repository includes a six-page author manuscript as `docs/source/files/colradpy_paper.pdf`.

[2] ADAS, *OPEN-ADAS report 1*, especially the generalized collisional radiative definitions and separate emission drivers. https://www.adas.ac.uk/adaseu_site/reports/open_1.pdf.

[3] ADAS User Manual, Appendix A, *ADF04: resolved specific ion data collections*. https://www.adas.ac.uk/man/appxa-04.pdf.

[4] ADAS User Manual, Appendix A, *ADF15: photon emissivity coefficients*. https://www.adas.ac.uk/man/appxa-15.pdf.

[5] ColRadPy development branch working tree used for this manual. The recorded HEAD, source-file hashes and runtime versions are in `docs/source/theory/figures/provenance.json`.

[6] NumPy developers, *NumPy 2.4.0 release notes*, expired deprecations and removed APIs. https://numpy.org/doc/2.4/release/2.4.0-notes.html.